

Aim: Implementation of Statistical Hypothesis Tests using Scipy and Scikit-learn.

Problem Statement: Perform the following Correlation Tests:

1. Pearson's Correlation Coefficient
2. Spearman's Rank Correlation
3. Kendall's Rank Correlation
4. Chi-Squared Test

Dataset Description:

The dataset consists of **614** loan applications with **13 columns** related to applicant details, income, credit history, and loan status. The dataset contains both numerical and categorical variables.

Column Name	Description	Data Type
Loan_ID	Unique Identifier for loan applications	Object
Gender	Gender of the applicant	Object
Married	Marital status of applicant	Object
Dependents	Number of dependents	Object
Education	Education level (Graduate/Not Graduate)	Object
Self_Employed	Employment status (Yes/No)	Object
ApplicantIncome	Applicant's monthly income	Integer
CoapplicantIncome	Co-applicant's monthly income	Float
LoanAmount	Loan amount requested	Float
Loan_Amount_Term	Term of loan in months	Float
Credit_History	Whether applicant has a credit history	Float
Property_Area	Area type (Urban/Semiurban/Rural)	Object
Loan_Status	Loan approval status (Y/N)	Object

Theory of Statistical Tests

1. Pearson's Correlation Coefficient

Pearson's correlation coefficient is a measure of the **linear relationship** between two continuous variables. It evaluates how well a straight line describes the relationship between two variables. It is given by:

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2} \sqrt{\sum(Y - \bar{Y})^2}}$$

where:

- X and Y are the two continuous variables.
- $\bar{X}$ and $\bar{Y}$ are their respective means.
- The numerator calculates the covariance between X and Y , while the denominator normalizes the value.
- **Range:** -1 to 1
- **Interpretation:**
 - $r = 1$: Perfect positive correlation (X increases, Y increases)
 - $r = -1$: Perfect negative correlation (X increases, Y decreases)
 - $r = 0$: No correlation

2. Spearman's Rank Correlation

Spearman's Rank Correlation assesses how well the relationship between two variables can be described using a **monotonic function**. Instead of using raw values, it ranks the data points and then calculates Pearson correlation on the ranks.

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where:

- d_i is the difference between the ranks of corresponding values.
- n is the number of observations.

Spearman's correlation is useful for non-linear relationships where data might be monotonically increasing or decreasing but not strictly linear.

3. Kendall's Rank Correlation

Kendall's Tau correlation coefficient measures the **ordinal association** between two variables. It is based on the number of concordant and discordant pairs.

$$\tau = \frac{C - D}{C + D}$$

where:

- **C** = Number of concordant pairs (where the order of **X** and **Y** is the same)
- **D** = Number of discordant pairs (where the order of **X** and **Y** is different)

Kendall's correlation is more robust for small datasets and is often used when data has tied ranks

4. Chi-Squared Test

The Chi-Squared test is a statistical method used to test the **independence** between two categorical variables. It compares the observed frequency of events with expected frequencies.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where:

- **O** = Observed Frequency (actual data counts)
- **E** = Expected Frequency (calculated assuming no relationship between variables)

This test helps determine whether two categorical variables (e.g., Education and Loan Approval) are dependent.

Python Libraries Used

1. **Pandas** - For handling dataframes
2. **Scipy.stats** - For performing statistical tests
3. **Numpy** - For mathematical operations

Output:

1] Correlation Matrix:-

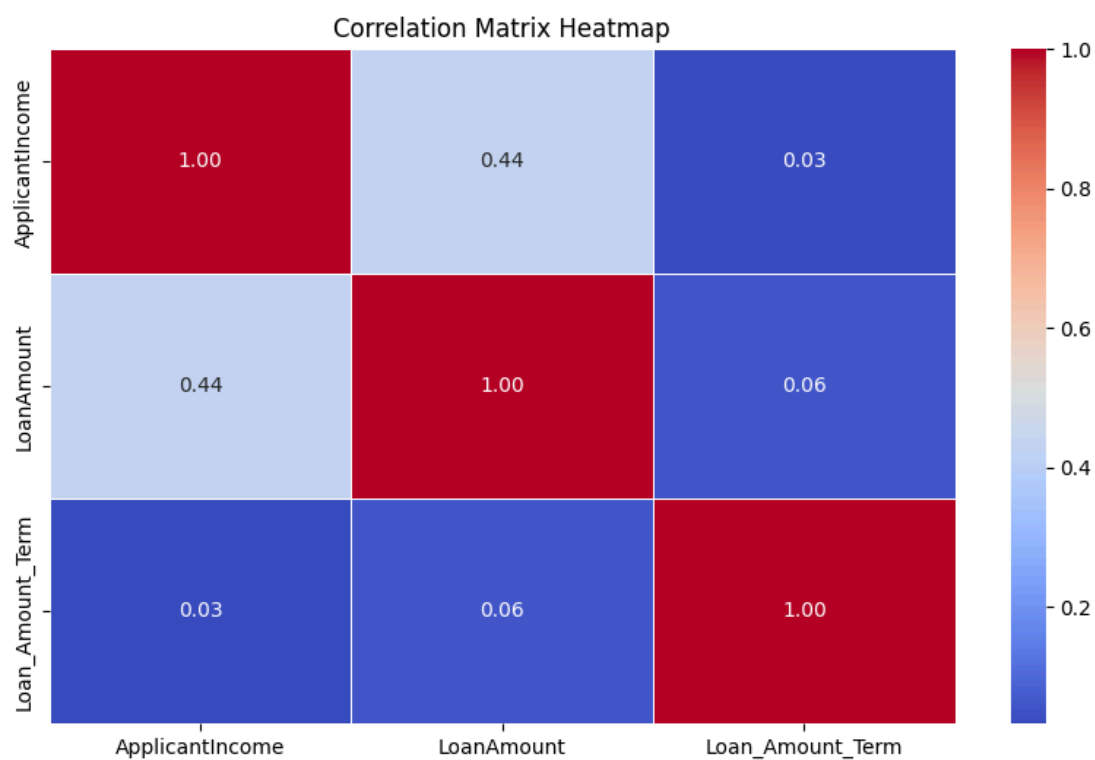
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Replace 0 with NaN and drop missing values
df.replace(0, pd.NA, inplace=True)
df_cleaned = df.dropna()

# Select only numeric columns
numeric_df = df_cleaned.select_dtypes(include=[np.number])

# Compute correlation matrix
correlation_matrix = numeric_df.corr()

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title("Correlation Matrix Heatmap")
plt.show()
```



Inference:-

a) ApplicantIncome and LoanAmount have a moderate positive correlation (0.44), meaning higher income applicants tend to request larger loans.

b) Loan_Amount_Term has almost no correlation with ApplicantIncome (0.03) and LoanAmount (0.06), indicating that loan term length does not depend much on income or loan amount.

2] Pearson's Correlation (Linear Relationship):-

```
[10] from scipy.stats import pearsonr

# Calculate Pearson's correlation between ApplicantIncome and LoanAmount
pearson_corr, p_value = pearsonr(df_cleaned['ApplicantIncome'], df_cleaned['LoanAmount'])

print(f"Pearson's Correlation: {pearson_corr:.4f}, p-value: {p_value:.4f}")
```

➔ Pearson's Correlation: 0.4352, p-value: 0.0000

A moderate positive correlation (0.4352) exists between ApplicantIncome and LoanAmount. Since the p-value is very small (< 0.05), the correlation is statistically significant. This means as ApplicantIncome increases, LoanAmount also tends to increase in a linear fashion.

3] Spearman's Correlation (Monotonic Relationship):-

```
[11] from scipy.stats import spearmanr

# Calculate Spearman's correlation
spearman_corr, p_value = spearmanr(df_cleaned['ApplicantIncome'], df_cleaned['LoanAmount'])

print(f"Spearman's Correlation: {spearman_corr:.4f}, p-value: {p_value:.4f}")
```

➔ Spearman's Correlation: 0.5542, p-value: 0.0000

Spearman's correlation is higher than Pearson's (0.5542 vs. 0.4352). This suggests a stronger monotonic relationship (i.e., even if the relationship isn't perfectly linear, higher income still tends to result in higher loan amounts). Again, the $p\text{-value} < 0.05$ confirms the relationship is significant.

4] Kendall's Rank Correlation (Ordinal Relationship):-

```
✓ [12] from scipy.stats import kendalltau
0s
# Calculate Kendall's correlation
kendall_corr, p_value = kendalltau(df_cleaned['ApplicantIncome'], df_cleaned['LoanAmount'])

print(f"Kendall's Correlation: {kendall_corr:.4f}, p-value: {p_value:.4f}")

⇒ Kendall's Correlation: 0.4046, p-value: 0.0000
```

The Kendall correlation is lower than Spearman but still positive (0.4046). This confirms a moderate positive association between ApplicantIncome and LoanAmount even when considering rank-based relationships. The p-value < 0.05 suggests this result is statistically significant.

5] Chi-Squared Test (Categorical Relationship):-

```
▶ from scipy.stats import chi2_contingency

# Creating a contingency table (Gender vs Loan Status)
contingency_table = pd.crosstab(df_cleaned['Gender'], df_cleaned['Loan_Status'])

# Perform Chi-Squared Test
chi2, p_value, dof, expected = chi2_contingency(contingency_table)

print(f"Chi-Squared Statistic: {chi2:.4f}, p-value: {p_value:.4f}")

⇒ Chi-Squared Statistic: 0.4225, p-value: 0.5157
```

The high p-value (> 0.05) suggests NO significant relationship between Gender and Loan Status. This means gender does not play a major role in whether a loan is approved.

Final Takeaways

1. **Strongest correlation: Spearman (0.5542)** → Suggests a **non-linear** but monotonic relationship between **income & loan amount**.
2. **Moderate correlation: Pearson (0.4352)** → Suggests a **linear** relationship.
3. **Weakest correlation: Chi-Squared (p = 0.5157)** → No significant link between **Gender & Loan Approval**.

Insights for Decision-Making

1. **Income plays a key role** in determining **Loan Amount**.
2. **Loan Amount is not strictly linearly dependent** on income but has a monotonic relationship.
3. **Gender does not significantly impact Loan Approval**.

Conclusion:

By implementing these statistical hypothesis tests, we:

- Measured the linear and monotonic relationships between numerical variables using Pearson, Spearman, and Kendall correlation tests.
- Tested categorical associations using the Chi-Squared test to check dependency between variables.
- Determine the statistical significance of these relationships with p-values.
- The results from these tests can be used for further predictive modeling and feature selection in machine learning applications.

Understanding correlations between features helps in **feature engineering**, **data preprocessing**, and **model interpretation** for improving predictions and decision-making processes.